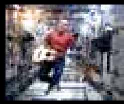
A Reddit AMA like no other

Here Cmdr Chris Hadfield replies to the many queries common people have about astronauts. This also happened to be one of the top Reddit AMA posts: **http://goo.gl/he2yK**



Space music video

This can safely be called the first ever music video recorded in space! Space Oddity by Cmdr Chris Hadfield **http://goo.gl/5Ay1J**

Eating

Food carried inside a space shuttle has to meet certain requirements. Carrying packaged food such as the type available in supermarkets is not an option, as the food needs to be stored for longer periods of time. Care also has to be taken that the food does not become a breeding ground for bacteria and fungi. Since moisture is the primary cause for micro-organism growth, the food packets found inside a space shuttle are partially or completely dehydrated. Meats are exposed to radiation to give them a higher shelf life. Forget using salt, pepper and other condiments in their natural state inside a spacecraft. In a microgravity environment crumbs and particles floating around can be hazardous.

Space food packets

So each food packet will have a dehydrated food with a small opening which goes into the water dispensing station which dispenses hot or cold water so that the food can be rehydrated.The ISS also has forced-air convection ovens to warm rehydratable, thermostabilized and irradiated food items. All drinks are had through a straw to prevent liquid from escaping.

Astronauts are given an eight-day menu decided by the space agency, which contains balanced meals with a calorie count of 1900 to 3200 calories per day depending on the weight and gender of the astronaut. Of course, the astronauts do get the option to carry some of their favourite food items which goes in the bonus containers, provided it is suitable inside a spacecraft.

As you must have realised by now, most of the food consumed on a spacecraft is sticky in nature. Also food aromas cannot really be smelt in zero gravity which in turn affects taste. Most food tastes bland - at least till the time the astronauts blood flow is regularised (as blood tends to be pushed to the brain which fills up the sinuses and affects sense of smell and taste).

Exercise

On earth we are always working against the gravity which keeps our muscles and bones in good condition as they are constantly being used. But in space in a zero gravity scenario, our bones and muscles aren't really being put into as much use. As a result your bone density drops and you tend to lose muscle mass as they are no longer employed to support your weight.

Exercising inside a space shuttle is not just a lifestyle decision, but mandatory to keep the bones and muscles in working condition. Astronauts have to be healthy to carry out work on the space station which includes spacewalks. On the ISS, the astronauts are expected to exercise for two hours daily and their exercise routine is monitored and can be adjusted according to data from daily exercise routine.

Astronauts use a treadmill (which has a harness system to secure the astronaut to the treadmill) for cardiovascular exercise, maintaining neuromuscular patterns for locomotion and a cycle ergometer which is like your stationary bicycle for cardiovascular exercise. Advanced Resistive Exercise Device (ARED) uses vacuum cylinders which apply loads of upto 600 pounds to allow astronauts to simulate weight-lifting exercises in space.

Advanced Resistive Exercise Device

An astronaut inside his sleeping bag

All the exercise equipment comes with a vibrations isolation system, to prevent astronauts who are working out from disturbing others on-board.

Some enterprising astronauts such as Sunita Williams have even completed a trialthon in space.

Sleeping

In zero gravity, where you float most of the time, sleeping can seem like a task. For starters, you have no need for mattresses or pillows here. On the International Space Station there are six sleep-pods, each of which contains a sleeping bag tied to the wall. All the astronaut has to do is get in them, zip up and float to sleep.

One thing to note with the ISS is that it witnesses a sunrise every 90 minutes, so the circadian rhythms that we are used to on earth, go for a toss in space. Also the constant whirring noise heard inside a space station can be distracting. They are provided with ear-plugs and eye-masks to drown out the noise. Most astronauts are used to taking sleep-promoting medications. Although 8-8.5 hours is the preferred sleep duration, most astronauts have reported that they just need 6-6.5 hours to feel fully rested.

NASA has a tradition of Wakeup Call songs: http://goo.gl/d5mCB

And just in case you were wondering, yes, it is possible to snore in space. d